Robotics and Machine Intelligence (WS 2026/27)

Micro-Quiz Practice Exercise Sheet

Artificial Intelligence for Programming (AIP) — Heidelberg University

Preparation for Quiz 1 (Lectures 1 & 2)

Quiz Format & Strategy (∼ 1 Minute Per Question)

The online micro-quiz consists of **3 multiple-choice questions** taken on `https://rmi-heidelberg.de/`. Once opened, you have strictly **1 minute per question**. These questions do *not* require lengthy computations: they test **core definitions, fundamental geometric transformations, matrix properties, and degrees of freedom (DoF)**.

Part 1: Quick-Fire Multiple-Choice Practice

Try to solve each of the following questions in under 60 seconds.

Q1. Forward vs. Inverse Kinematics
Which of the following statements is **TRUE**?

(a) Forward kinematics maps gripper poses in task space to joint angles.
(b) For any open serial robot arm, the forward kinematics solution is unique.
(c) Inverse kinematics always yields exactly one unique solution for any serial arm.
(d) Forward kinematics is generally computationally harder than inverse kinematics.

Q2. Properties of Rotation Matrices in SO(3)
Let $R \in \mathbb{R}^{3\times3}$ be a valid rotation matrix. Which of the following conditions must hold?

(a) $R^{-1} = -R^\top$ and $\det(R) = 1$
(b) $RR^\top = I$ and $\det(R) = -1$
(c) $R^{-1} = R^\top$ and $\det(R) = 1$
(d) $\|Rv\| = 0$ for any unit vector v

Q3. Degrees of Freedom (DoF)
How many degrees of freedom does an unconstrained rigid body have in 2D space and in 3D space, respectively?

(a) 2 DoF in 2D space, 3 DoF in 3D space
(b) 3 DoF in 2D space, 6 DoF in 3D space
(c) 3 DoF in 2D space, 7 DoF in 3D space
(d) 6 DoF in 2D space, 6 DoF in 3D space

Q4. The Inverse of a Homogeneous Transformation Matrix
Given the homogeneous transformation matrix $T_{sb} = \begin{pmatrix} R & p \\ 0 & 1 \end{pmatrix} \in \mathrm{SE}(3)$, where $R \in \mathrm{SO}(3)$ and $p \in \mathbb{R}^3$, what is its inverse $T_{sb}^{-1} = T_{bs}$?

(a) $\begin{pmatrix} R^\top & -p \\ 0 & 1 \end{pmatrix}$

(b) $\begin{pmatrix} R^\top & -R^\top p \\ 0 & 1 \end{pmatrix}$

(c) $\begin{pmatrix} -R^\top & p \\ 0 & 1 \end{pmatrix}$

(d) $\begin{pmatrix} R & -Rp \\ 0 & 1 \end{pmatrix}$

Q5. Order of Rotations in 3D

A rigid body is rotated by 90° around the fixed space axis $\hat{x}_s$ ($R_1 = \text{Rot}(\hat{x}, 90°)$), and then subsequently rotated by 90° around the fixed space axis $\hat{z}_s$ ($R_2 = \text{Rot}(\hat{z}, 90°)$). What is the total composite rotation matrix R_{total}?

(a) $R_{total} = R_1 R_2$

(b) $R_{total} = R_2 R_1$

(c) $R_{total} = R_1 + R_2$

(d) $R_{total} = R_1 R_2 = R_2 R_1$ because rotations always commute

Q6. Reading a Rotation Matrix

Consider the rotation matrix $R_{sb} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Along which axis of the fixed space frame $\{s\}$ does the body frame axis $\hat{x}_b$ point?

(a) Along $+\hat{x}_s$

(b) Along $-\hat{y}_s$

(c) Along $+\hat{y}_s$

(d) Along $+\hat{z}_s$

Q7. Orientation Representations and Gimbal Lock

Which representation of 3D orientations suffers from the singularity known as *Gimbal Lock*?

(a) Unit quaternions

(b) Roll-Pitch-Yaw / Euler angles

(c) 3×3 Rotation matrices

(d) Axis-Angle representation

Q8. Point Transformation with Homogeneous Coordinates

Let point $p_b = (1, 2, 0)^\top$ be given in body frame $\{b\}$. Frame $\{b\}$ is translated by $p_{sb} = (3, 0, 0)^\top$ relative to $\{s\}$ with no rotation ($R_{sb} = I$). What is the representation of the point in space frame $\{s\}$?

(a) $p_s = (3, 0, 0)^\top$

(b) $p_s = (4, 2, 0)^\top$

(c) $p_s = (-2, 2, 0)^\top$

(d) $p_s = (1, 2, 0)^\top$

Q9. Joints and Redundancy

The course robot arm (Annin AR4) has 6 revolute joints. A research arm like the Franka Emika Panda has 7 revolute joints. Why is the 7-DoF arm called *redundant* for positioning and orienting a tool in 3D space?

(a) It has more than 6 DoF, meaning it can reach a target 6D pose with infinitely many internal arm configurations.

(b) It has spare backup motors in case one joint fails.

(c) It cannot reach all 6 dimensions of task space.

(d) Its forward kinematics cannot be written using matrix multiplication.

Q10. Chaining Transformations

Suppose we know T_{sc} (camera $\{c\}$ relative to base $\{s\}$) and T_{co} (object $\{o\}$ relative to camera $\{c\}$). We want to find the pose of the object relative to the gripper $\{e\}$, denoted T_{eo}. We also know the current forward kinematics of the gripper T_{se}. What is the correct formula for T_{eo}?

(a) $T_{eo} = T_{se}T_{sc}T_{co}$

(b) $T_{eo} = T_{se}^{-1}T_{sc}T_{co}$

(c) $T_{eo} = T_{sc}T_{co}T_{se}^{-1}$

(d) $T_{eo} = T_{so}T_{se}$

Part 2: Solutions & Detailed Explanations

Solutions Key

Q1: (b) **Q2:** (c) **Q3:** (b) **Q4:** (b) **Q5:** (b)
Q6: (c) **Q7:** (b) **Q8:** (b) **Q9:** (a) **Q10:** (b)

Detailed Explanations

- **Q1: (b)** For open-chain serial manipulators, forward kinematics is a deterministic chain of matrix products $T_{01}(\theta_1)\cdots T_{nb}$; given θ, there is always **exactly one** pose. In contrast, inverse kinematics can have multiple solutions (elbow-up vs. elbow-down) or no solution (outside reachable workspace).
- **Q2: (c)** By definition of the Special Orthogonal group SO(n): $RR^\top = I \implies R^{-1} = R^\top$, and the "Special" condition requires $\det(R) = +1$. A determinant of -1 indicates an improper rotation (reflection).
- **Q3: (b)** In a plane (2D), a rigid body has 3 DoF: (x, y, θ). In 3D space, a rigid body requires 6 numbers to describe its pose: 3 for translation (x, y, z) and 3 for orientation (e.g., roll, pitch, yaw).
- **Q4: (b)** Because $T = \begin{pmatrix} R & p \\ 0 & 1 \end{pmatrix}$, multiplying by $\begin{pmatrix} R^\top & -R^\top p \\ 0 & 1 \end{pmatrix}$ gives:

$$\begin{pmatrix} R & p \\ 0 & 1 \end{pmatrix}\begin{pmatrix} R^\top & -R^\top p \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} RR^\top & -RR^\top p + p \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} I & -p+p \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & 1 \end{pmatrix} = I$$

  Common exam trap: forgetting to multiply p by $-R^\top$.
- **Q5: (b)** Rotations about **fixed axes** (the room frame $\{s\}$) are composed by multiplying consecutive rotations from the **left**: $R_{total} = R_2 R_1 = \text{Rot}(\hat{z}, 90°)\text{Rot}(\hat{x}, 90°)$. (If rotations were taken about moving body axes, they would multiply from the right).
- **Q6: (c)** The i-th column of R_{sb} represents the i-th basis vector of frame $\{b\}$ expressed in frame $\{s\}$. The first column is $\hat{x}_b = (0, 1, 0)^\top$, which is directly along the $+\hat{y}_s$ axis.
- **Q7: (b)** Euler angles (such as Roll-Pitch-Yaw) suffer from **gimbal lock** whenever the middle pitch angle reaches $\pm 90°$. At that singularity, the first and third axes align, collapsing two rotational degrees of freedom into one. Quaternions and rotation matrices do not suffer from gimbal lock.
- **Q8: (b)** For point transformation in homogeneous coordinates: $p_s = R_{sb}p_b + p_{sb} = I\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix}$.
- **Q9: (a)** Tool poses in 3D task space require 6 DoF (3 positions, 3 orientations). An arm with ≥ 7 DoF has extra internal degrees of freedom: with the gripper pinned in place, the elbow can still trace an arc without moving the end-effector.
- **Q10: (b)** Using the subscript cancellation rule: First, find the object in base frame: $T_{so} = T_{sc}T_{co}$. Then, express the object relative to the gripper: $T_{eo} = T_{es}T_{so} = T_{se}^{-1}(T_{sc}T_{co})$.